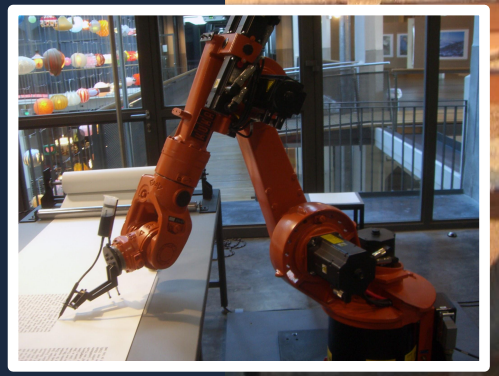
Fast and Smooth: Trajectory Optimization for a 3-DOF Robot Arm

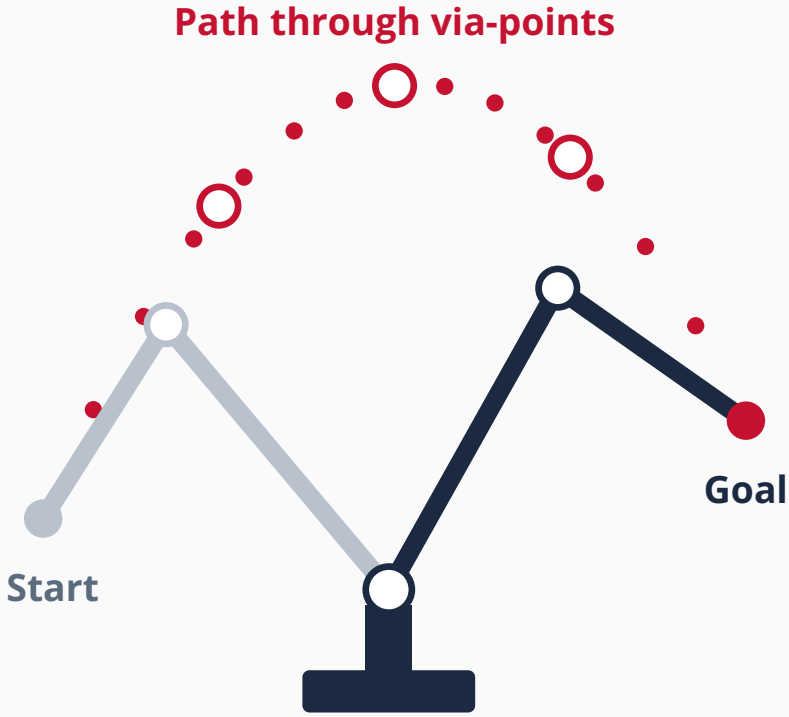
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Goal

Find the **fastest smooth motion** for a 3-DOF arm between a fixed start and goal pose.

- **Fast:** minimize total motion time
- **Smooth:** minimize joint jerk
- **Feasible:** within physical limits



Objective Function

Minimize **total motion time** plus a weighted penalty on the **squared jerk** of all three joints, integrated over the whole motion.

Compact form

$$J = \underbrace{T}_{\text{speed}} + \underbrace{w \sum_{i=1}^{3} \int_0^T j_i(t)^2 \, dt}_{\text{smoothness}}$$

Expanded over the three joints

$$J = T + w \left[\int_0^T j_1^2 \, dt + \int_0^T j_2^2 \, dt + \int_0^T j_3^2 \, dt \right]$$

Computed form, in decision variables

$$\min_{\mathbf{q},\,\Delta\mathbf{t}} \; J(\mathbf{q}, \Delta\mathbf{t}) = \sum_{k=1}^{4} \Delta t_k + 36\, w \sum_{i=1}^{3} \sum_{k=1}^{4} d_{ik}(\mathbf{q}, \Delta\mathbf{t})^2 \, \Delta t_k$$

$$d_{ik}(\mathbf{q}, \Delta\mathbf{t}) = \frac{v_{i,k-1} + v_{ik}}{\Delta t_k^2} - \frac{2\left(q_i^{(k)} - q_i^{(k-1)}\right)}{\Delta t_k^3}$$

Where

q, Δt: the 13 decision variables

$q_i^{(k)}$: angle of joint i at k

Δt_k: duration of segment k

v_{ik}: velocity at knot k (from spline)

d_{ik}: cubic term, jerk = $6d_{ik}$

$j_i = d^3q_i/dt^3$: jerk of joint i

$T = \Sigma\Delta t_k$: total motion time

w: smoothness weight

i, k: joint index, segment index

Red: decision variables

Constraints

Boundary Conditions

$$q_i(0) = q_{i,s}, \quad q_i(T) = q_{i,g}$$

$$\dot{q}_i(0) = \dot{q}_i(T) = 0$$

Kinematic Constraints

$$\dot{\mathbf{q}} = \frac{d\mathbf{q}}{dt}, \qquad \ddot{\mathbf{q}} = \frac{d^2\mathbf{q}}{dt^2}$$

$$q_{i,\min} \leq q_i(t) \leq q_{i,\max}$$

$$|\dot{q}_i(t)| \leq \dot{q}_{i,\max}, \qquad |\ddot{q}_i(t)| \leq \ddot{q}_{i,\max}$$

Dynamic Constraints

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) = \boldsymbol{\tau}$$

$$|\tau_i(t)| \leq \tau_{i,\max}$$

Smoothness Constraints

$$j_i(t) = \frac{d^3 q_i}{dt^3}, \qquad |j_i(t)| \leq j_{i,\max}$$

Time and Spline Constraints

$$\Delta t_k > 0, \qquad \sum_{k=1}^{n} \Delta t_k = T$$

Solving Method

1 Define Variables

Decision variables:

$q_{i,k}$ and Δt_k

Compute $d_{i,k}$:

$$v_{i,k-1} + 4v_{i,k} + v_{i,k+1} = \frac{3}{h}\left(q_i^{(k+1)} - q_i^{(k-1)}\right)$$

$$d_{ik}(\mathbf{q}, \Delta\mathbf{t}) = \frac{v_{i,k-1} + v_{ik}}{\Delta t_k^2} - \frac{2\left(q_i^{(k)} - q_i^{(k-1)}\right)}{\Delta t_k^3}$$

2 Formulate NLP

Implement objective and constraints in Python

$$J(\mathbf{q}, \Delta\mathbf{t}) = \sum_{k=1}^{4} \Delta t_k + 36\, w \sum_{i=1}^{3} \sum_{k=1}^{4} d_{ik}(\mathbf{q}, \Delta\mathbf{t})^2\, \Delta t_k$$

Constraints: joint, velocity, jerk and torque limits.

3 Optimize

Use an NLP solver

e.g., SciPy SLSQP or IPOPT

Find optimal joint angles and segment durations.

Output: optimized trajectory and total time.

4 Validate

Load trajectory into Simulation

Check tracking and torque limits

Verify smoothness and feasibility.

Compare simulated motion with constraints.

The optimal trajectory is expected to balance speed and smoothness

Assumptions

- Rigid 3-DOF manipulator
- Smooth cubic spline trajectories
- Fixed start and end poses
- No obstacle avoidance

Expected Insights

Small *w*	**Large *w***
Faster motion	Smoother motion
Higher Jerk	Lower Jerk
More active constraints	Less active constraints

- Different trajectories may satisfy the same task, potentially leading to multiple optima

Stretch Goals

1. Animate the optimized trajectory
2. Add obstacles to avoid
3. Simulate PID tracking of the optimized trajectory

Questions?